

Yusuf Naeem Abd El-Hakim

SOFTWARE ENGINEER

Cairo, Egypt · +20 102 090 0480 · yusufnaemhakim@gmail.com · binhakim.dev · GitHub · LinkedIn

SUMMARY

Software Engineer who takes systems from a blank schema to production without supervision. Three years designing REST APIs, relational data models, and access-control layers built to hold under concurrent load, alongside two self-directed SaaS products taken solo from idea to public launch. Owns what comes after launch too: reliability, performance, and security hardening, with offensive-security training in the OWASP Top 10 shaping how systems are threat-modelled before they ship rather than patched afterwards. Equally comfortable on the commercial side, gathering requirements across business units, running cost-benefit analysis for executive decisions, and redesigning the processes the software is meant to serve.

EDUCATION

B.Sc. Computer Science and Information Technology, Software Engineering Major

Ahram Canadian University · GPA 3.36 / 4.00 · Final Semester 4.00 · Graduation Project A+

Sep 2019 – Jul 2023

Activities: ACU Competitive Programming Community · CyberTalents

Volunteering: Peer tutoring for classmates in Algorithms, Software Engineering, Design Patterns, and Information Security.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Backend: Node.js, Express.js, REST API Design, PostgreSQL, Supabase

Frontend: React, Next.js, Three.js, D3, HTML, CSS

AI / LLMs: Gemini API Integration, Semantic Search, Prompt Engineering

Infrastructure: Docker, Nginx, Vercel, AWS (S3, EC2), CI/CD

Tools: Git, Claude Code, Antigravity, Postman, Swagger, Jira

Practices: Agile / Scrum, TDD, BDD (Cucumber)

Also worked with: Java / Spring Boot, Django REST Framework, Power BI, OWASP Top 10, Burp Suite

PROFESSIONAL EXPERIENCE

Software Engineer | EPPS Egyptian Packaging & Plastic Systems, Cairo · Hybrid · Full-time since Jun 2024

Apr 2024 – Present

- Built and operate an internal B2B lead pipeline platform replacing spreadsheet tracking across four business entities, holding roughly 2,000 records for the 8 to 10 staff running marketing through to closed sale.
- Designed a full audit trail and an atomic identifier generator that keeps records conflict-free under concurrent writes.
- Resolved a Marketing and Sales data-ownership conflict with field-level, server-side access control that strips restricted fields from API payloads before the client sees them.
- Built SC 3D Optimizer, applying a 3D bin-packing heuristic to container load planning under door-clearance constraints, cutting planning time fourfold and improving space utilisation (React, Three.js).
- Modelled the data and lifecycle workflows for the IT asset management platform, tracking roughly 3,000 assets with full check-out and check-in history.
- Extended the company's public web presence with new pages and features, taking each from business requirement to production deployment.
- Integrated analytics and tracking tools across the sites, instrumenting performance measurement to guide optimisation.
- Own reliability and security across 25+ domains and 7 production sites through caching, automated backups, and web server hardening.

Offensive Security Engineer | WE INNOVATE Cairo, Egypt · Hybrid · Internship

Sep 2024 – Dec 2024

- Exploited web, mobile, and infrastructure vulnerabilities against the OWASP Top 10, documenting each with a reproducible proof-of-concept and remediation path (Burp Suite, Metasploit, Wireshark).
- Ran Active Directory and network penetration tests, mapping privilege-escalation and lateral-movement paths across the target environment.
- Wrote Python tooling automating recurring reconnaissance and vulnerability-assessment workflows, beyond the internship's defined scope.
- Reviewed application source and configuration for insecure defaults in authentication, session handling, and input validation, returning each as a hardening change.

Software Engineer | HomeBay Furniture Ontario, Canada · Remote

Nov 2023 – Oct 2024

- Automated product catalogue and asset synchronisation through the Google Sheets API, removing a recurring manual task from the team's weekly cycle (Python).
- Extended WooCommerce product-management and checkout functionality, applying caching, database indexing, and image optimisation across the storefront.

- Built new storefront pages and features, and implemented on-page SEO across the catalogue to improve search visibility.
- Owned day-to-day site reliability, monitoring traffic and error rates to catch issues before customers did.

Software Engineer | Kites Software *Cairo, Egypt · Hybrid · Internship*

Aug 2023 – Nov 2023

- Shipped RESTful API endpoints test-first for Egypt's first SaaS ERP for import and distribution, with BDD integration tests in Cucumber (Java, Spring Boot, PostgreSQL).
- Diagnosed and fixed QA-reported defects inside a full Agile and Scrum cycle.
- Authored onboarding documentation that the team adopted for subsequent intern cohorts.

Business Analyst | Tact Decoration *Cairo, Egypt · Remote · Part-time*

Jun 2023 – Aug 2023

- Led the company's first digital-transformation initiative, replacing untracked, informal coordination with a central system.
- Built a CRM and operations hub in Notion from scratch, giving site engineers, architects, and supervisors one platform for task assignment and tracking.
- Redesigned cross-functional workflows between field and office teams, documenting the new process as a standard other staff could follow.

Freelance Full-Stack Developer *Remote · Self-employed*

Jan 2022 – Jun 2022

- Shipped websites and online storefronts for small businesses end to end, from static frontend (HTML, CSS, JavaScript) through backend logic and data (Django, DRF).
- Worked directly with each business owner to scope requirements and deliver a working site without a team or supervision.
- Implemented storefront features including product catalogues and inquiry forms, backed by Django REST Framework APIs.

SELECTED PROJECTS & PRODUCTS

Commit | Curriculum Tracker for Self-Taught Developers

commit.binhakim.dev

Sole Designer and Engineer · Launched May 2026 · Node.js, PostgreSQL, SQLite, Resend

- Designed a git-conceptualised learning model mapping study sessions to commits, streaks to contribution graphs, and completions to merges, turning self-study into a measurable engagement loop.
- Built a database-agnostic data access layer allowing a single-line swap between local SQLite and production PostgreSQL, eliminating environment drift and reducing setup to one command.
- Implemented an ETA engine projecting completion dates from observed study pace, alongside roadmap visualisation, XP milestones, and automated daily digests (Resend).
- Took the product from personal tool to public launch: scoped requirements, designed the schema, implemented the API, and now operate it in production.

Orbit | Professional Relationship Operating System

byorbit.io

Sole Designer and Engineer · Live, in active development · Next.js, TypeScript, Supabase (PostgreSQL), Gemini, Stripe

- Enforced tenant isolation at the database layer with PostgreSQL row-level security, so isolation holds even when application code is wrong.
- Implemented Google OAuth and magic-link authentication, and Stripe Free/Pro billing with usage metering.
- Integrated the Gemini API for semantic search across a user's connections, and built a D3 force-directed graph for exploring network structure.
- Shipped full English and Arabic right-to-left localisation, packaged as an installable Progressive Web App.

iREAD | Reading Platform and Digital Bookstore

2023

Graduation Project, Ahram Canadian University · Supervised by Dr. Sherif El-Shafey, Head of Software Engineering · Flutter, React, REST API

- Built a reading platform combining a catalogue with reviews and reading history, a storefront for digital purchases, and a community layer for exchanging insights.
- Designed a REST API serving both the Flutter mobile and React web clients, keeping one source of business logic across platforms.

PUBLICATIONS

Technical Articles | Medium (@binhakim)

Aug 2025 – Oct 2025

- Published articles on backend engineering, database design, and web security for practitioner and general audiences.

CERTIFICATIONS

- CS50: Introduction to Computer Science, HarvardX
- Web Development and .NET Development Summer Training, Information Technology Institute (ITI)
- Offensive Security Engineering, Certificate of Completion, WE INNOVATE
- Forward Program, McKinsey.org

LANGUAGES

- Arabic (native or bilingual)
- English (professional working)
- German (limited working)